

INTRODUCTION TO CREATING A CHATBOT

BASIC AI CHATBOT

GROQ API AND RAG SYSTEM

OVERVIEW

This project focuses on building AI-powered conversational systems using Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG).

The first part involves developing a chatbot using the Groq API and Llama models for multi-turn conversation.

The second part implements a RAG pipeline that retrieves relevant information from PDF documents and generates context-aware answers.

Together, these projects demonstrate the practical application of modern AI technologies

PURPOSE

The purpose of this project is to build an intelligent AI system that provides accurate, context-aware, and document-based responses using LLM and RAG technologies.

It focuses on solving the following problems:

- ✦ Traditional chatbots often provide generic or inaccurate responses without verifying information from reliable sources.
- ✦ Searching through large documents manually is time-consuming and inefficient for users.
- ✦ Existing systems lack context-aware retrieval, making it difficult to generate precise answers from specific documents or knowledge sources.

WHAT IS GROQ API?



GROQ API LETS YOUR PROGRAM TALK TO AN AI MODEL AND GET SMART ANSWERS.

- Groq API provides access to powerful AI language models through simple API requests.
- It processes user input and generates intelligent text-based responses in real time.
- It is optimized for very fast inference, making chatbot interactions quicker.
- Developers can integrate it into applications such as chatbots, RAG systems, and AI assistants.

TERMS RELATED TO GROQ API

1. **Multi-turn conversation** – The chatbot remembers previous messages to continue a natural conversation.
2. **Temperature** – A setting that controls how creative or random the AI responses are.
3. **Max tokens** – The maximum length of the response generated by the AI model.
4. **Chat behaviour** – The way the chatbot responds based on instructions or system prompts given to it.

FEATURES OF GROQ API

1

Multi-turn
conversation
support

2

Conversation
memory for
maintaining
context

3

Adjustable
temperature and
max token
settings

4

Prompts for
controlling
chatbot
behavior

WHAT IS RAG?

RAG IS A METHOD WHERE THE AI SEARCHES DOCUMENTS FOR INFORMATION AND THEN USES IT TO GENERATE ACCURATE ANSWERS.

- RAG stands for retrieval-augmented generation
- RAG combines document retrieval with AI-generated responses.
- It searches relevant documents before answering the user's question.
- It helps the AI provide more accurate and context-based answers.
- It is useful for applications like chatbots, document analysis, and knowledge assistants.

TERMS RELATED TO RAG

1. **Semantic search** – A search method that finds information based on meaning instead of exact keywords.
2. **Embeddings** – Numerical representations of text that help the AI understand similarities between words and sentences.
3. **ChromaDB** – A vector database used to store and retrieve document embeddings efficiently.
4. **Vector database integration** – The process of connecting embeddings with ChromaDB to enable fast semantic document retrieval.

FEATURES OF RAG

1

PDF document
loading and
processing

2

Semantic search
using
embeddings

3

ChromaDB
vector database
integration

4

Answer
generation using
retrieved
documents

CONTRIBUTIONS

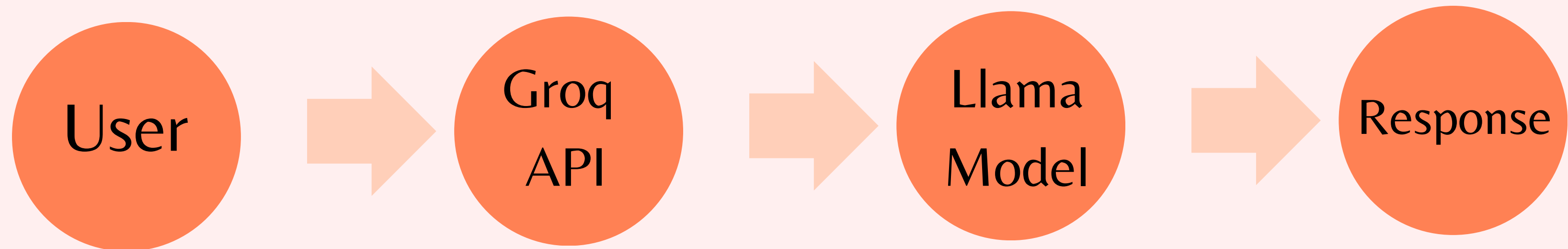
- Designed and developed both the LLM chatbot and RAG-based document retrieval system using Python.
- Integrated Groq API, Llama models, LangChain, HuggingFace embeddings, and ChromaDB into the projects.
- Implemented features such as multi-turn conversation, conversation memory, semantic search, document chunking, and context-aware response generation.
- Handled project setup, dependency management, debugging, testing, GitHub repository management, and documentation preparation.

CHALLENGES FACED

- Understanding and configuring the Groq API and selecting a compatible LLM model.
- Learning concepts such as embeddings, semantic search, and vector databases used in RAG systems.
- Extracting clean text from PDF documents due to formatting and spacing inconsistencies.
- Choosing appropriate chunk sizes and overlap values to improve retrieval accuracy.
- Managing library dependencies, environment setup, and resolving installation-related errors during development.

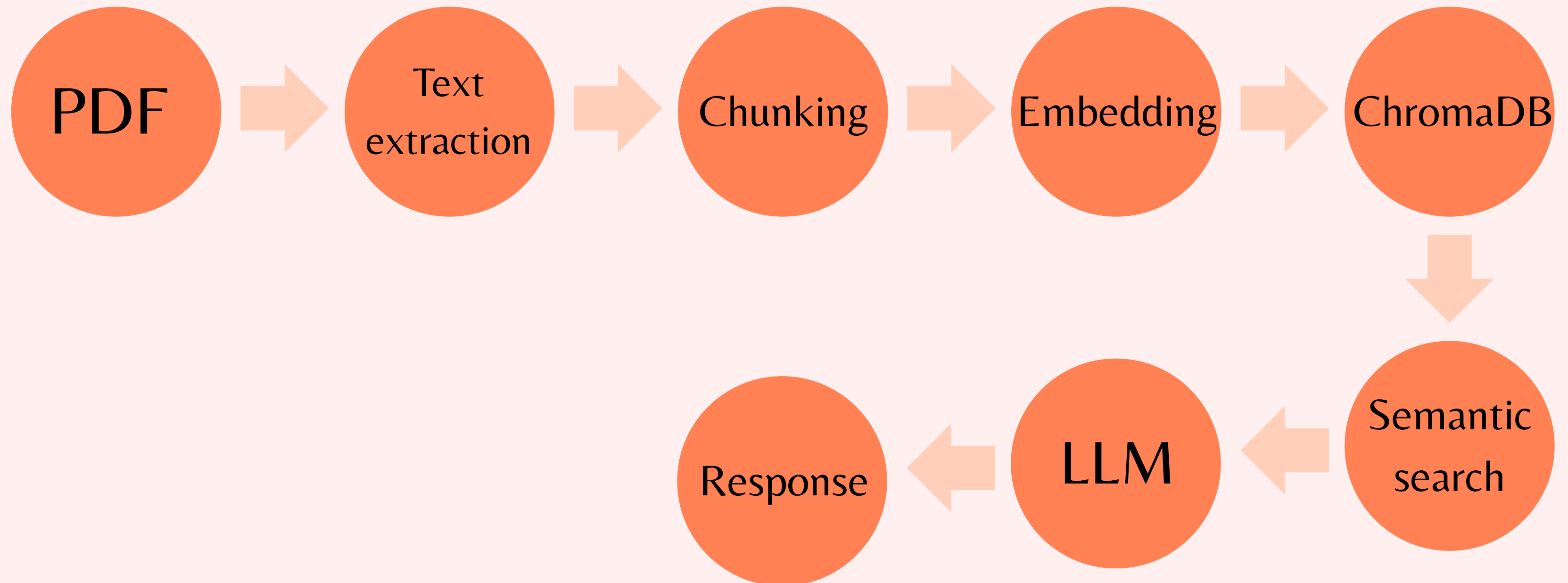
SYSTEM ARCHITECTURE

For chatbot:



SYSTEM ARCHITECTURE

For RAG:



EXECUTION

To run the project, the required dependencies must first be installed and the Groq API key must be configured. Once the environment is set up, the application can be executed through the terminal using Python. For the LLM chatbot, users can interact directly with the AI through a conversational interface, while the RAG system additionally loads and processes PDF documents, retrieves relevant information using semantic search, and generates context-aware responses. The system continues accepting user queries until the exit command is entered, providing an interactive and efficient AI-powered experience.

CONCLUSION

This project demonstrates the successful development of an intelligent AI-powered solution that combines conversational capabilities with knowledge retrieval. By integrating Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG), the system delivers faster, more accurate, and context-aware responses. The solution reduces the effort required to search through large volumes of information, improves user experience, and highlights the potential of AI-driven systems.

Overall, the project showcases how modern AI technologies can be leveraged to create efficient, scalable, and reliable information assistance solutions.

THANK YOU

